Name: _____

Class: X Section: A

Subject: Biology Date: 7.08.2026

Chapter/Topic: Control and Coordination

Delhi Public School
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Nurturing the Whole Child

Q1. Choose and tick the most appropriate answer –

1. [CBSE Board] Which of the following statements is correct about the propagation of a nerve impulse in a neuron?

(a) Chemical signal travels from dendritic end towards axon end.
(b) Electrical impulse travels from dendritic end towards axonal end, and at the synaptic terminal, it releases neurotransmitter chemicals.
(c) Chemical signal travels from cell body to dendritic end.
(d) Electrical signal travels from axon end to cell body across the synaptic cleft.

2. [CBSE Board] In a neuron, conversion of an electrical signal to a chemical signal occurs at/in:

(a) Cell body (Cyton)
(b) Axonal end
(c) Dendritic end
(d) Axon midpoint

3. [CBSE Board] Which part of the human brain is responsible for maintaining posture and balance of the body, as well as precision of voluntary actions like walking in a straight line?

(a) Cerebrum
(b) Cerebellum
(c) Medulla oblongata
(d) Pons

4. [CBSE Sample Paper] Electrical impulse travels in a neuron from:

(a) Dendrite → Axon → Axonal end → Cell body
(b) Cell body → Dendrite → Axon → Axonal end
(c) Dendrite → Cell body → Axon → Axonal end
(d) Axonal end → Axon → Cell body → Dendrite

5. [CBSE Board] Posture and balance of the body are controlled by the cerebellum, whereas involuntary actions such as blood pressure, salivation, and vomiting are controlled by the:

(a) Cerebrum in the forebrain
(b) Medulla in the hindbrain
(c) Pons in the hindbrain
(d) Spinal cord independently

6. [CBSE Board] Which of the following is a plant hormone that inhibits growth and causes wilting of leaves during environmental stress?

(a) Auxin
(b) Gibberellin
(c) Cytokinin
(d) Abscisic acid

7. [CBSE Board] The movement of sunflower leaves in response to sunlight and the movement of a tendril around a support are examples of:

(a) Both are nastic movements.
(b) Sunflower movement is nastic, while tendril movement is tropic.
(c) Both are tropic movements.
(d) Sunflower movement is tropic, while tendril movement is nastic.

8. [CBSE Sample Paper] When a potted plant is kept horizontally near an open window, its stem bends upwards towards light, and roots grow downwards into the soil. What types of tropisms are demonstrated here?

(a) Positively geotropic stem and negatively phototropic root.
(b) Positively phototropic & negatively geotropic stem; negatively phototropic & positively geotropic root.
(c) Negatively phototropic stem and positively geotropic root only.
(d) Nastic movement in both stem and root systems.

9. [CBSE Board] A person eats food deficient in Iodine for a prolonged period. Which endocrine gland will be affected, and what condition is likely to develop?

(a) Pancreas → Diabetes mellitus
(b) Thyroid gland → Goitre
(c) Adrenal gland → Addison's disease
(d) Pituitary gland → Acromegaly

10. [CBSE Sample Paper] Dwarfism results from the hypo-secretion of Growth Hormone. Which gland secretes this hormone?

(a) Thyroid gland
(b) Pituitary gland
(c) Adrenal gland
(d) Hypothalamus

11. A patient involved in an accident suffers a head injury. Upon clinical examination, the patient can understand spoken language and remember past events, but loses the ability to coordinate precise hand movements required to button a shirt. Which specific brain region was most likely damaged?

(a) Cerebrum
(b) Cerebellum
(c) Hypothalamus
(d) Medulla oblongata

12. A seedling is grown inside a dark chamber with a single slit emitting light from the right side. The stem bends towards the light source. What happens to the concentration of Auxin on the shaded (left) side of the shoot tip?

(a) Auxin concentration decreases on the shaded side, slowing down cell growth.

(b) Auxin concentration increases on the shaded side, stimulating greater cell elongation compared to the illuminated side.
(c) Auxin degrades completely on the shaded side due to lack of light.
(d) Auxin diffuses evenly, causing uniform stem growth on both sides.

13. A person experiences a sudden spike in blood glucose levels after a meal. Which response sequence correctly describes the feedback control mechanism activated in the human body?

(a) Liver detects high glucose → secretes glucagon → cells absorb excess sugar.
(b) Pancreatic cells detect high glucose → secrete insulin → liver/muscle cells take up glucose to store as glycogen.
(c) Pituitary gland detects high glucose → secretes thyroxin → kidney excretes sugar.
(d) Adrenal gland detects high glucose → secretes adrenaline → accelerates cellular respiration.

14. Which of the following correct sequences represents the reflex arc when a person accidentally steps on a sharp pin?

(a) Effector (Leg muscle) → Motor neuron → Spinal cord → Sensory neuron → Receptor (Skin)
(b) Receptor (Skin) → Sensory neuron → Spinal cord (Relay neuron) → Motor neuron → Effector (Leg muscle)
(c) Receptor (Skin) → Motor neuron → Brain → Sensory neuron → Effector (Leg muscle)
(d) Sensory neuron → Receptor (Skin) → Spinal cord → Motor neuron → Effector (Leg muscle)

15. Which plant hormone is present in high concentration in rapidly dividing plant tissues such as seeds and fruits, promoting active cell division?

(a) Abscisic acid
(b) Gibberellin
(c) Cytokinin
(d) Ethylene

Q2. Assertion & Reasoning -

Directions: In each of the following questions, a statement of **Assertion (A)** is given, followed by a corresponding statement of **Reason (R)**. Select the correct option:

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(b) Both (A) and (R) are true, but (R) is NOT the correct explanation of (A).

(c) (A) is true, but (R) is false.

(d) (A) is false, but (R) is true.

1. [CBSE Board]

- **Assertion (A):** The movement of leaves of the sensitive plant (*Mimosa pudica*) in response to touch is independent of plant growth.
- **Reason (R):** Plant cells alter their shape by changing the amount of water (turgor pressure) in them, resulting in swelling or shrinking.

2. [CBSE Board]

- **Assertion (A):** Impulse transmission across a chemical synapse is unidirectional.
- **Reason (R):** Neurotransmitter chemicals are stored and released only from axon terminals, and their specific receptors are present only on the dendritic membranes of the next neuron.

3. [CBSE Board]

- **Assertion (A):** Iodised salt is recommended in our daily diet.
- **Reason (R):** Iodine is essential for the synthesis of thyroxin hormone by the thyroid gland, which regulates carbohydrate, protein, and fat metabolism in the body.

4. [CBSE Sample Paper]

- **Assertion (A):** Tendrils of a pea plant coil around any support they come in contact with.
- **Reason (R):** The part of the tendril in direct contact with the support grows faster than the part away from the support due to rapid cell division.

5. [CBSE Board]

- **Assertion (A):** Reflex actions are controlled and mediated primarily at the level of the spinal cord without immediate brain processing.
- **Reason (R):** Spinal reflexes provide an extremely rapid survival response to dangerous stimuli before the brain can process sensory inputs.

6. [CBSE Board]

- **Assertion (A):** Adrenaline is called the "emergency hormone" or "fight-or-flight hormone".
- **Reason (R):** Adrenaline secretion increases heart rate, breathing rate, and blood pressure while diverting blood supply to skeletal muscles during stressful situations.

7.

- **Assertion (A):** Damage to the medulla oblongata leads to immediate stoppage of breathing and heart function, resulting in fatal outcomes.
- **Reason (R):** Medulla oblongata contains vital autonomic reflex centers that control involuntary functions like respiration, heartbeat, and blood pressure.

8.

- **Assertion (A):** Plants lack a specialized nervous system and muscle tissue, yet they can sense environmental stimuli and respond directionally.
- **Reason (R):** Plants use electrical-chemical means to transfer information from cell to cell and alter cellular water content or cell elongation rates.

9.

- **Assertion (A):** In a diabetic patient, insulin injections are administered subcutaneously to lower elevated blood glucose levels.
- **Reason (R):** Insulin is produced by the alpha cells of the thyroid gland to convert glycogen into glucose.

10.

- **Assertion (A):** Phototropism in plant shoots is brought about by the uneven distribution of Auxin.
- **Reason (R):** Auxin synthesised at the shoot tip diffuses toward the illuminated side, causing cells on the light-exposed side to grow longer than those on the shaded side.